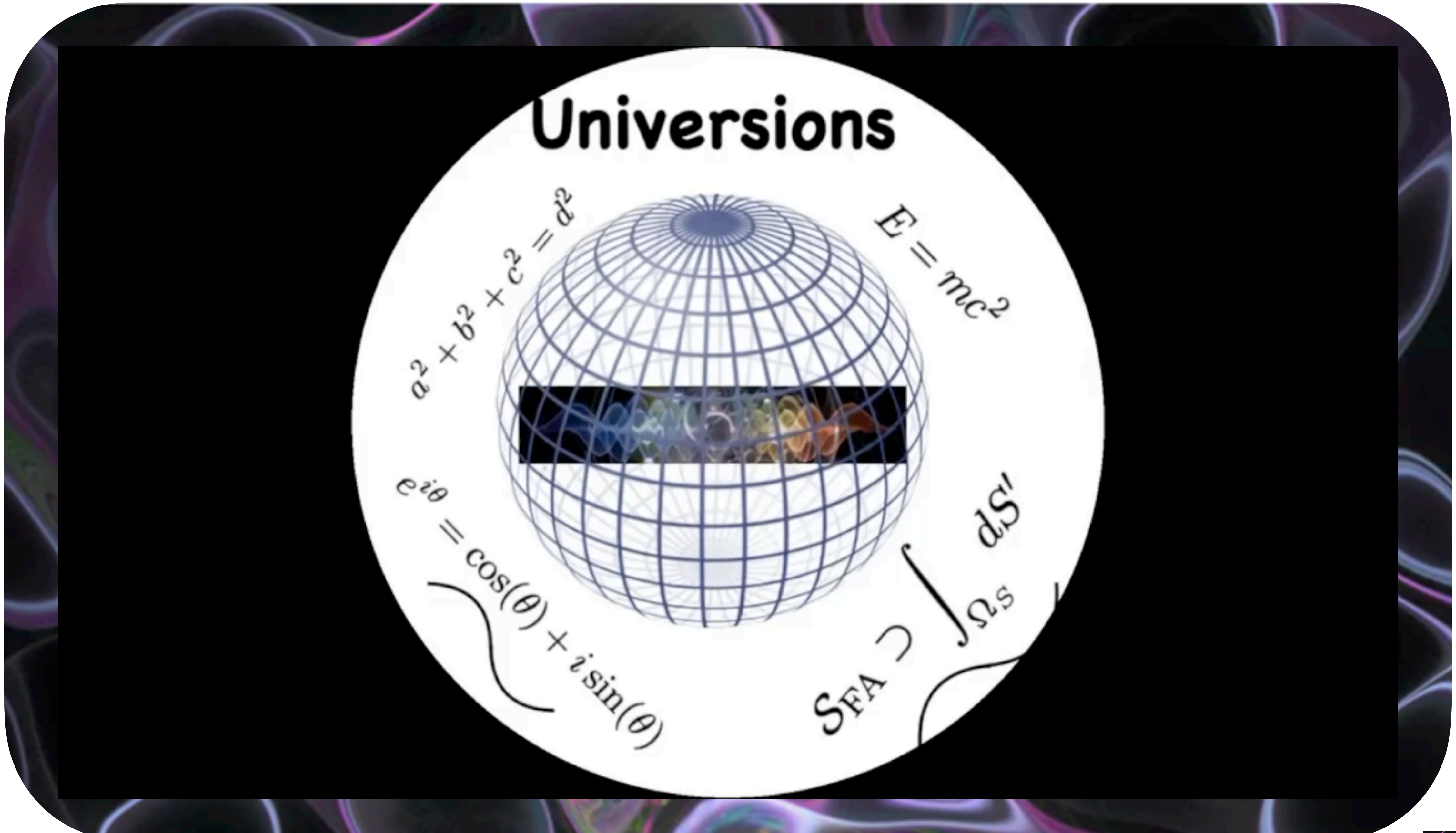
Institute of Theoretical Physics
National University of Uzbekistan

A cosmic whirlwind

哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

Analytical polyΛCDM in cosmic observations



Prof. Pierros Ntelis

July 2026

Universions

Outline

Introduction

Theoretical framework

- polyΛCDM (in cosmic observations)

Recent results

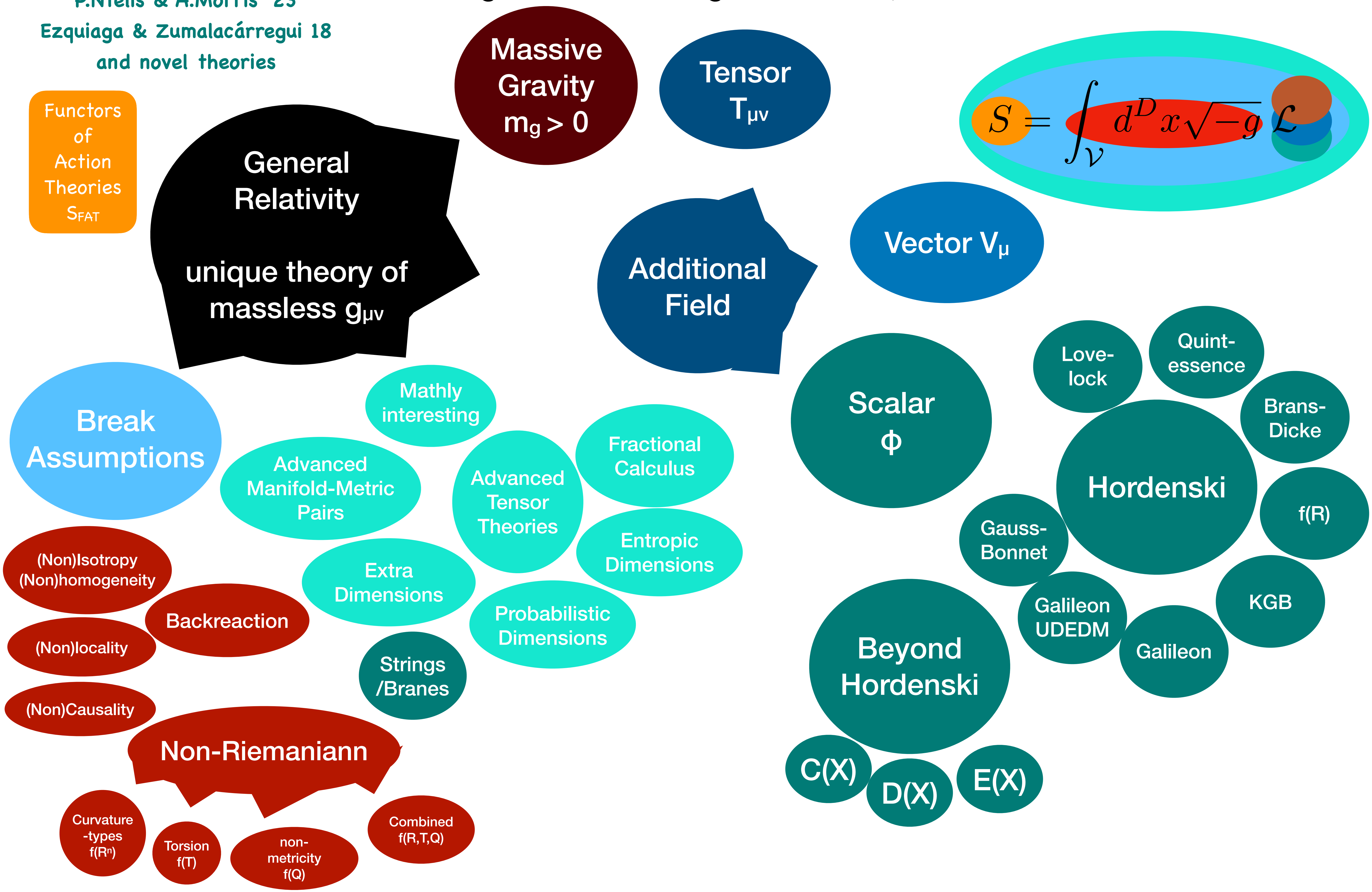
- Functors of Actions Theories (FAT) (Cosmos APP)
- Quintessence
- Extra Dimensions in Observations
- Advanced Manifold-Metrics Pairs (AMMP)
- A temporal-spatial-entropic K curvature (EinsteinKPy2Tex code)
- Advanced Tensor Theories (ATT)
- A Foundational Analysis of Kernel-based Calculus (AFAKEC)

- A cosmic femtolensing Hawking radiation (Poster)

Conclusions and future

Diagram remodified from
P.Ntelis & A.Morris '23
Ezquiaga & Zumalacárregui 18
and novel theories

COsmological GRAvitological TheORy (COGRATORY)



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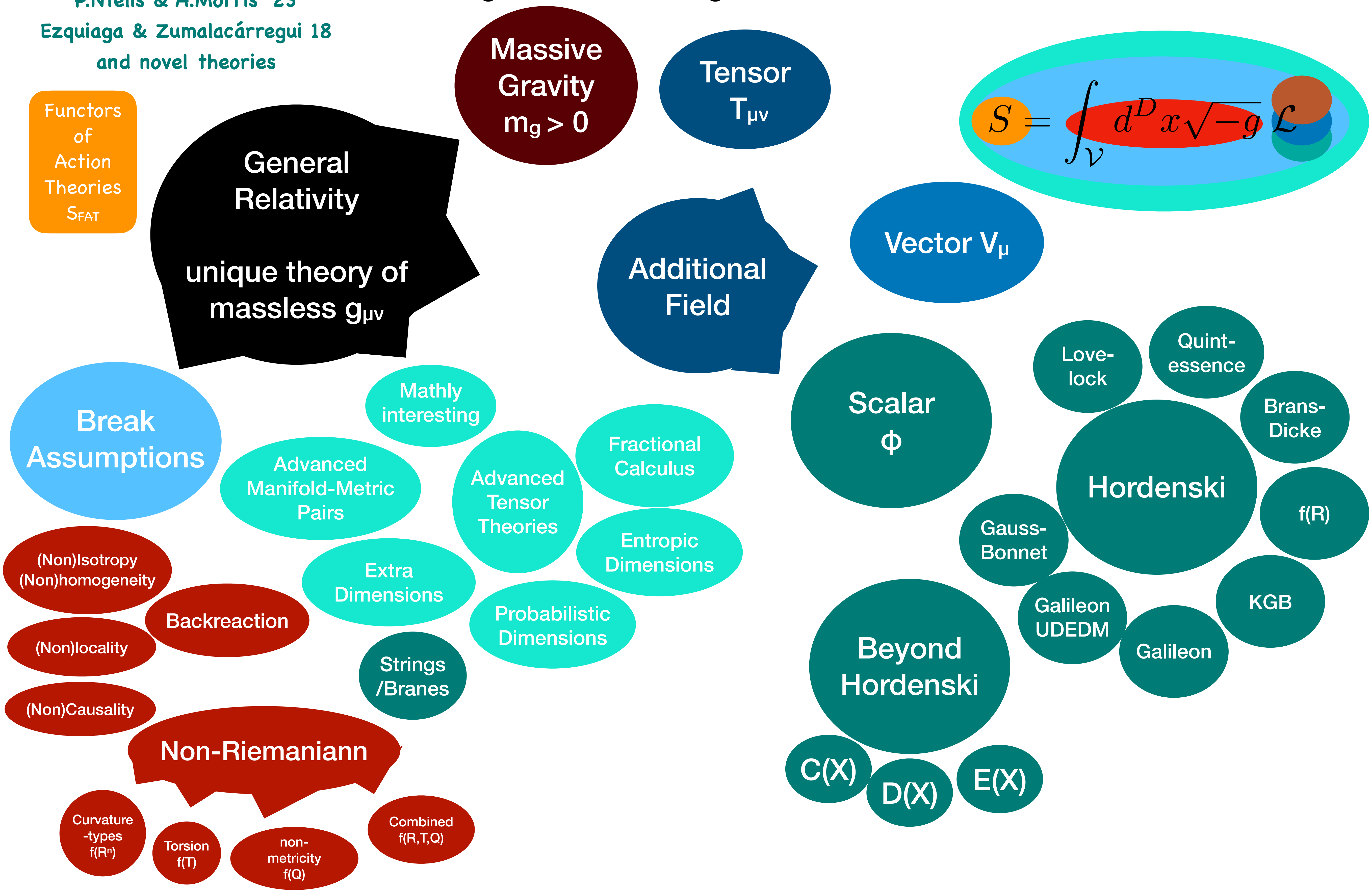

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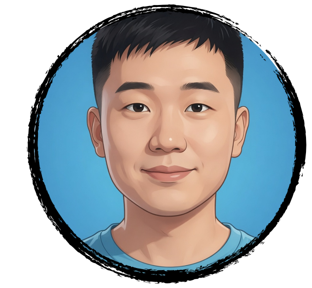
Conclusions and future

Which effective model can help us distinguish among Modified Gravity Cosmological Models ?

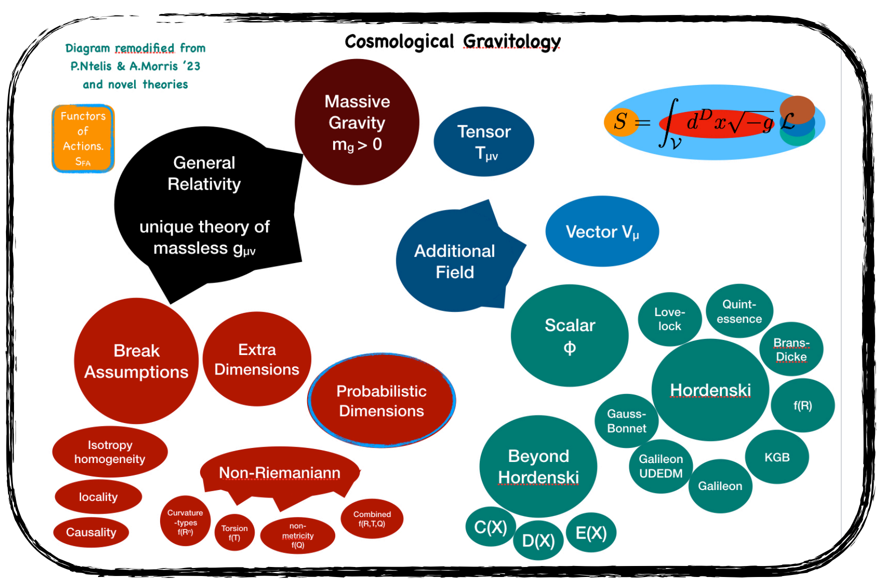


ZX & PN under prep.

Analytical *poly*ΛCDM dynamics

Pierros Ntelis[a], Jackson Levi Said[b,c]

[a] *Independent Research Affiliation Former Aix Marseille Univ. Marseille France*
[b] *Institute of Space Sciences and Astronomy University of Malta Msida Malta*
[c] *Department of Physics University of Malta Msida Malta*



... Further Simulations ...

Standard ΛCDM + Modified Gravity and DE

Or in details

L = (m, r, Λ) + (IDE, f(R), Curvature, SVT)

$$H(a) = \sum_{s\in\{m,r,\Lambda,x,k,z,v\}} \Omega_{s,0} a^{-\alpha_s}$$

From Action to Effective Lagrangians which model Modified Gravity models

$$
\begin{aligned}
H(z) = H_0 [\quad & \Omega_{m,0}(1+z)^3 && \text{Matter} \\
& +\Omega_{r,0}(1+z)^4 && \text{Radiation} \\
& +\Omega_{x,0}(1+z) && \text{f(R)} \\
& +\Omega_{k,0}(1+z)^2 && \text{Curvature} \\
& +\Omega_{z,0}(1+z)^5 && \text{Interacting Dark Energy} \\
& +\Omega_{v,0}(1+z)^\alpha && \text{SVT} \\
& +\Omega_{\Lambda,0} && \text{Cosmo Constant} \\
& \quad]^{1/2}
\end{aligned}
$$

Analytical polyΛCDM dynamics

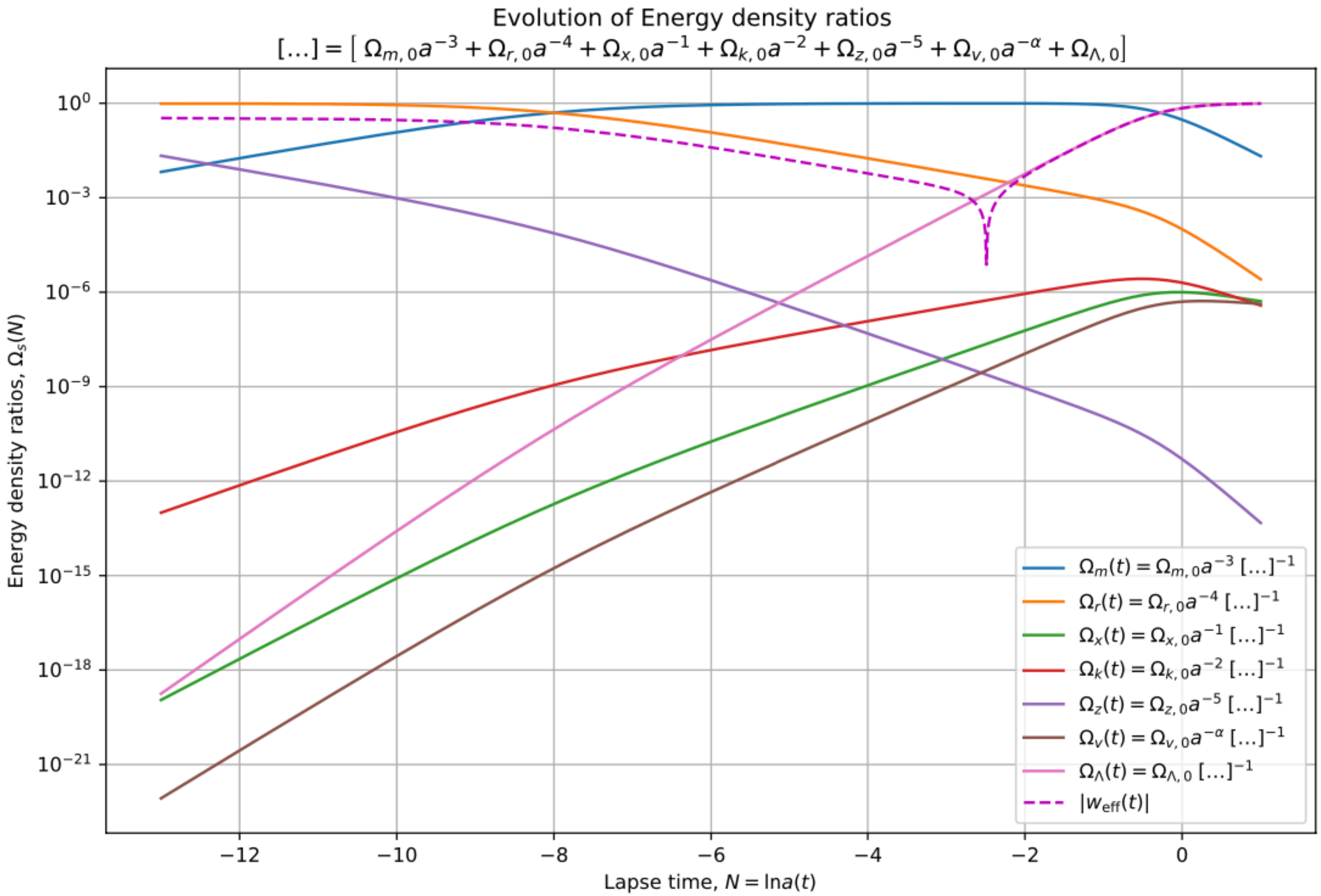


Figure 5: Energy density ratio of different species, Ω_s versus lapse time, N, of *poly*ΛCDM. [See section 3.3.2]

Analytical polyΛCDM dynamics

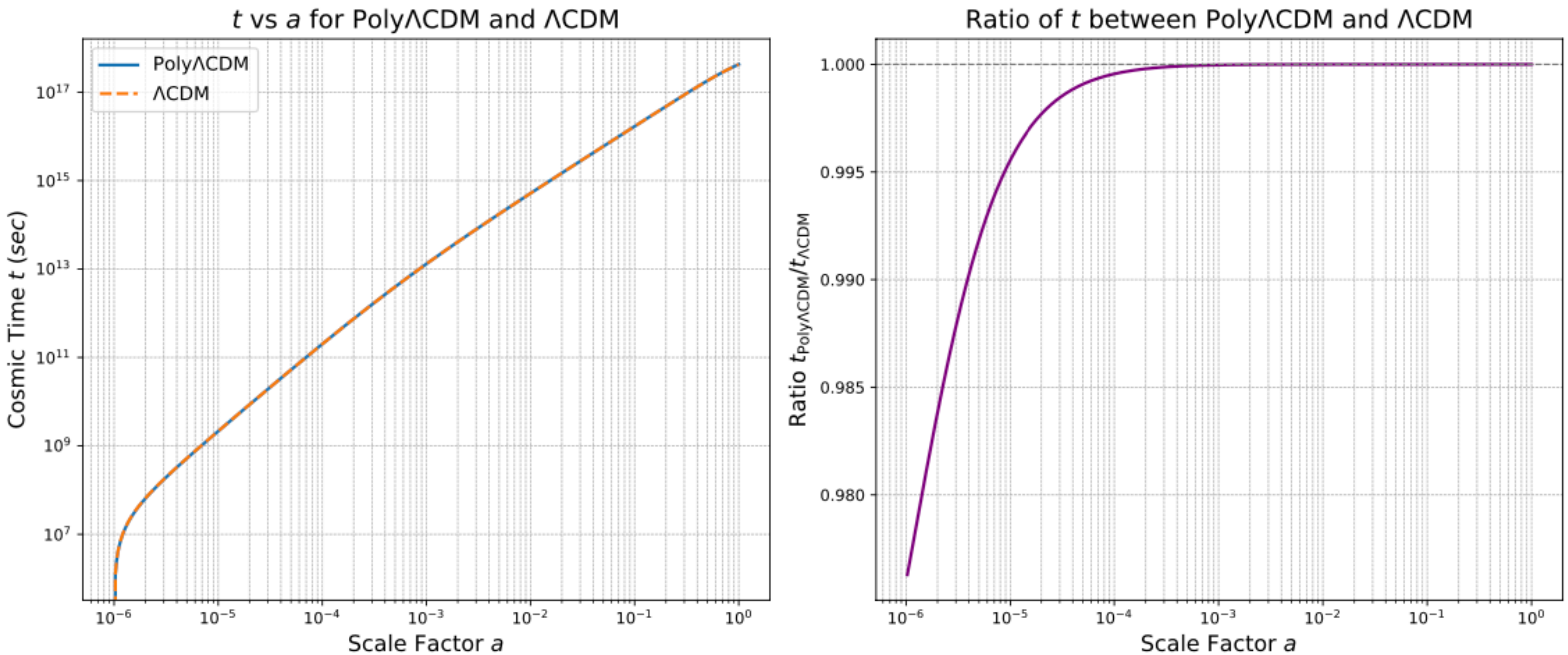


Figure 6: Left: Time, t versus scale factor, a, for the 2nd Friedman equation, of *poly*ΛCDM and of ΛCDM. Right: Ratio of time, t, between *poly*ΛCDM and ΛCDM versus scale factor, a. [See section 3.3.3]

for *poly*ΛCDM cosmology is

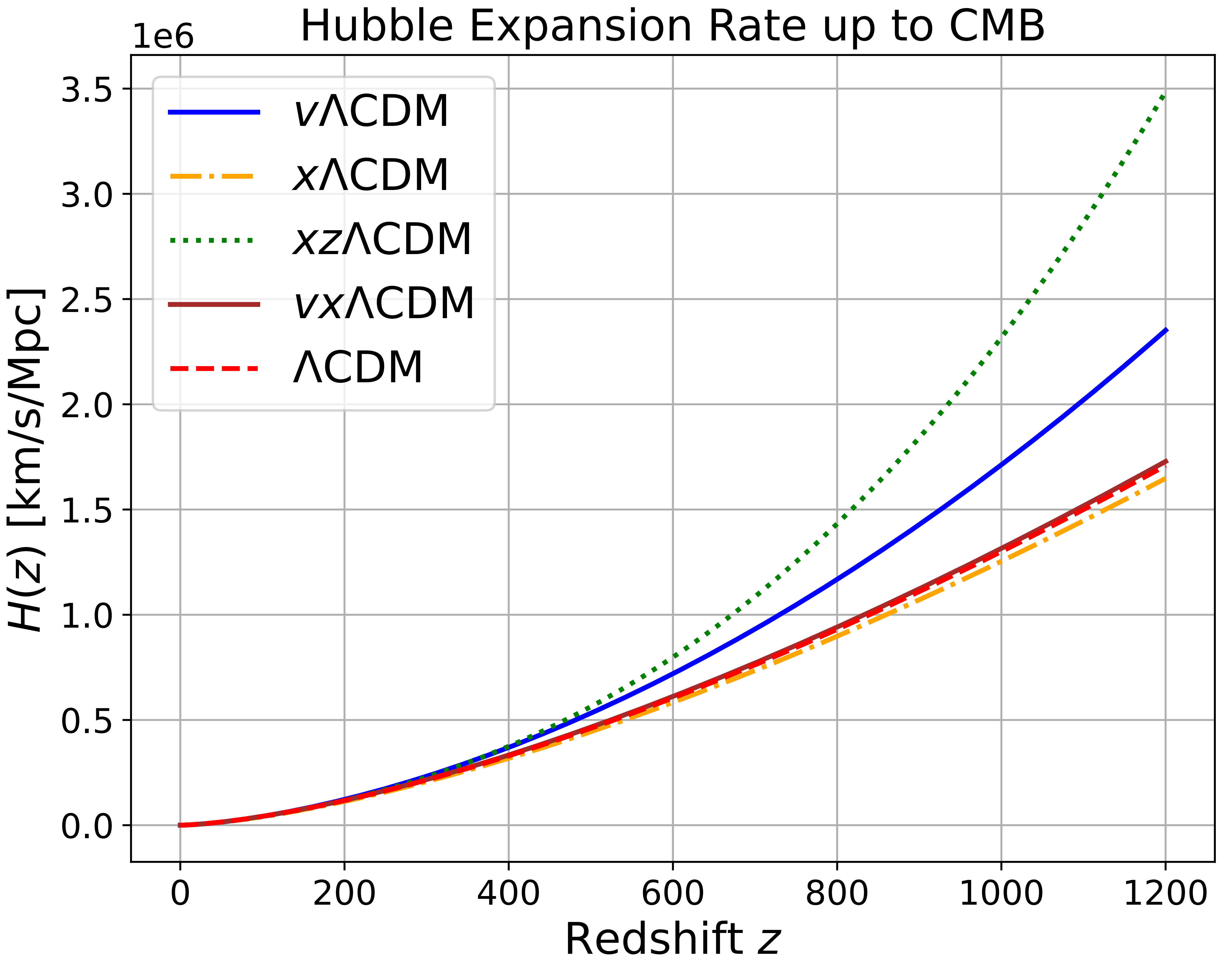
$$\boxed{t - t_0 = H_0^{-1} \int_{a_0}^{a} \frac{da}{a\left[\sum_s \Omega_{s,0} a^{-\alpha_s}\right]^{\frac{1}{2}}}}$$

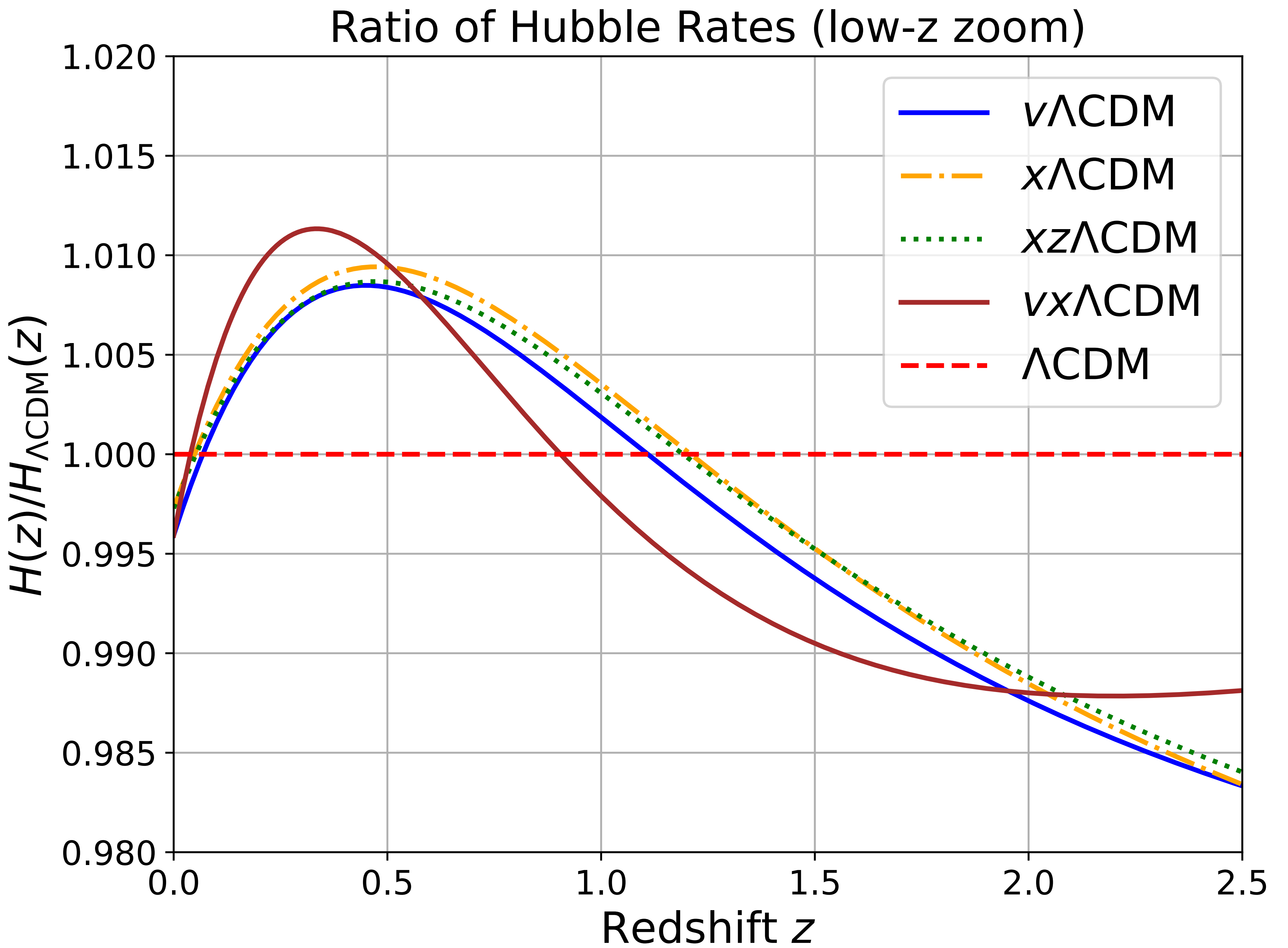
ΛCDM cosmology, which is

$$\boxed{t - t_0 = H_0^{-1} \int_{a_0}^{a} \frac{da}{a\left[\Omega_{m,0} a^{-3} + \Omega_{r,0} a^{-4} + \Omega_{\Lambda,0}\right]^{\frac{1}{2}}}} \quad (67)$$

where $a_0 = 10^{-6}$.

Potential Observational Constrains from H(z): BAO: dA(z), dH(z), and SNe: dL(z)


Hubble Expansion Rate up to CMB
1e6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$H(z)$ [km/s/Mpc]
0
200
400
600
800
1000
1200
Redshift z
$v\Lambda$CDM
$x\Lambda$CDM
$xz\Lambda$CDM
$vx\Lambda$CDM
ΛCDM


Ratio of Hubble Rates (low-z zoom)
1.020
1.015
1.010
1.005
1.000
0.995
0.990
0.985
0.980
$H(z)/H_{\Lambda\mathrm{CDM}}(z)$
0.0
0.5
1.0
1.5
2.0
2.5
Redshift z
$v\Lambda$CDM
$x\Lambda$CDM
$xz\Lambda$CDM
$vx\Lambda$CDM
ΛCDM

Analytical polyΛCDM dynamics

$$\chi^2_{\rm total} = \chi^2_{\rm SNIa} + \chi^2_{\rm BAO} + \chi^2_{\rm CMB}$$

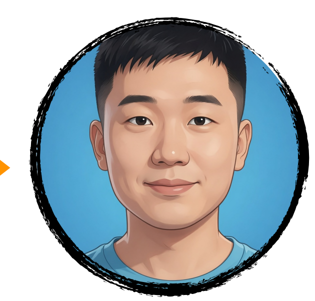
TABLE III. Model comparison statistics. ΔAIC and ΔAICc are defined as ΛCDM minus the alternative model, so positive values favour the alternative. The guidelines for interpreting $|\Delta|$ are given in Appendix A.

Model	$\chi^2_{\rm min}$	k	AIC	AICc	ΔAIC	ΔAICc
ΛCDM	1538.53	5	1548.53	1548.60	0.00	0.00
$v\Lambda$CDM	1535.49	6	1547.49	1547.53	1.04	1.07
$x\Lambda$CDM	1535.66	6	1547.66	1547.70	0.87	0.90
$xz\Lambda$CDM	1535.68	7	1549.68	1549.73	–1.15	–1.13
$vx\Lambda$CDM	1534.66	7	1548.66	1548.71	–0.13	–0.11

Better performance

Worse performance

ΛCDM still remains, but **Update** DATA/Method ...



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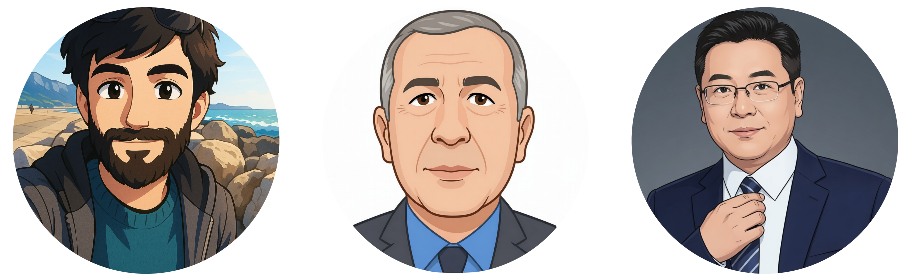
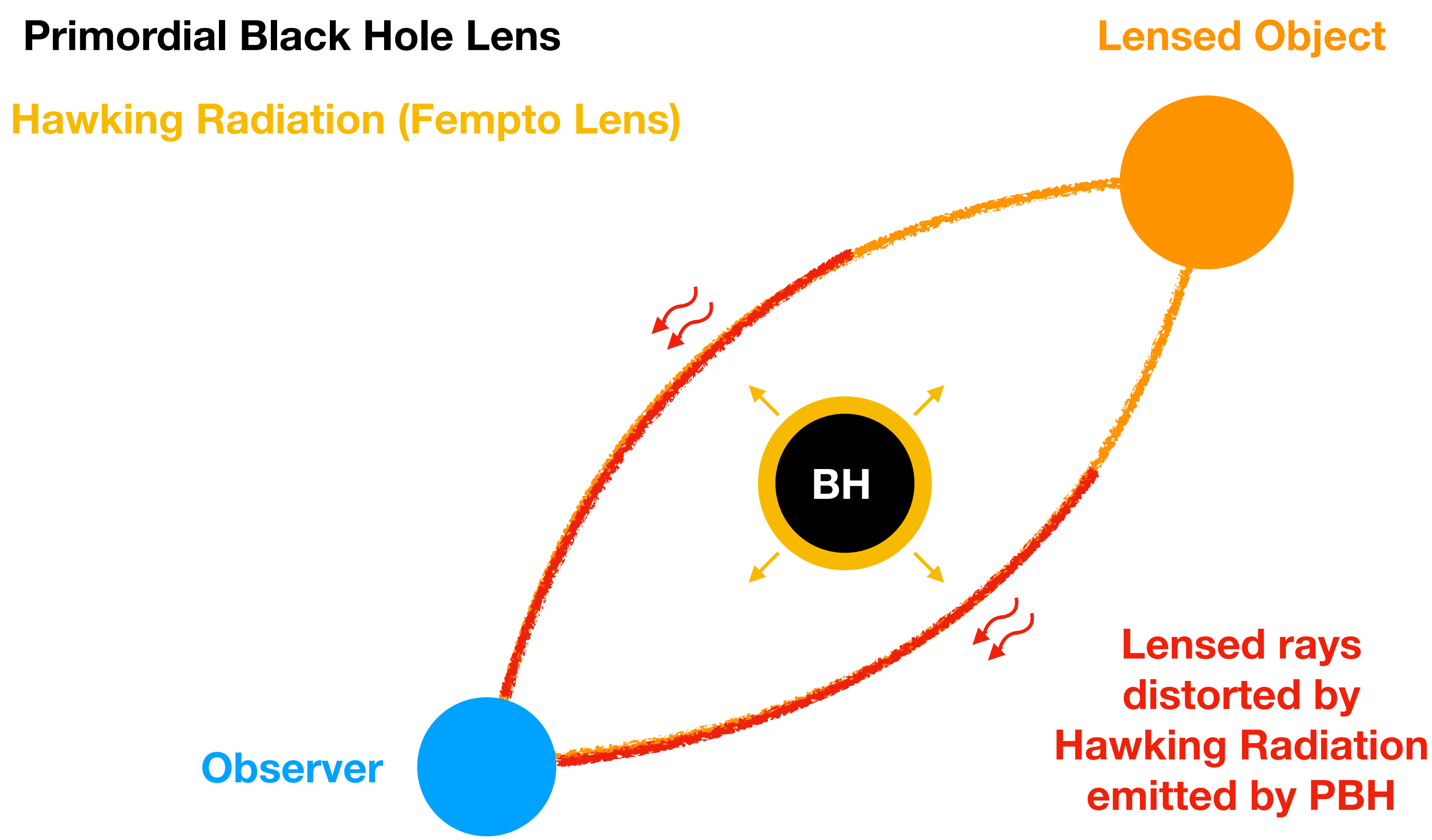
A cosmic femtolensing Hawking radiation (Poster)

PN, B.J. Ahmedov, C. Yuan





Thank you

谢谢

Xiè xiè

Ευχαριστώ

! :)

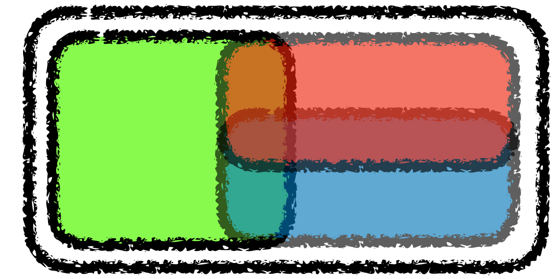
Functors of actions

Pierros Ntelis[1, a)] and Adam Morris[2, b)]
[1)] *Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France*
[2)] *Universität Bonn - Helmholtz-Institut für Strahlen und Kernphysik, Bonn, Germany*

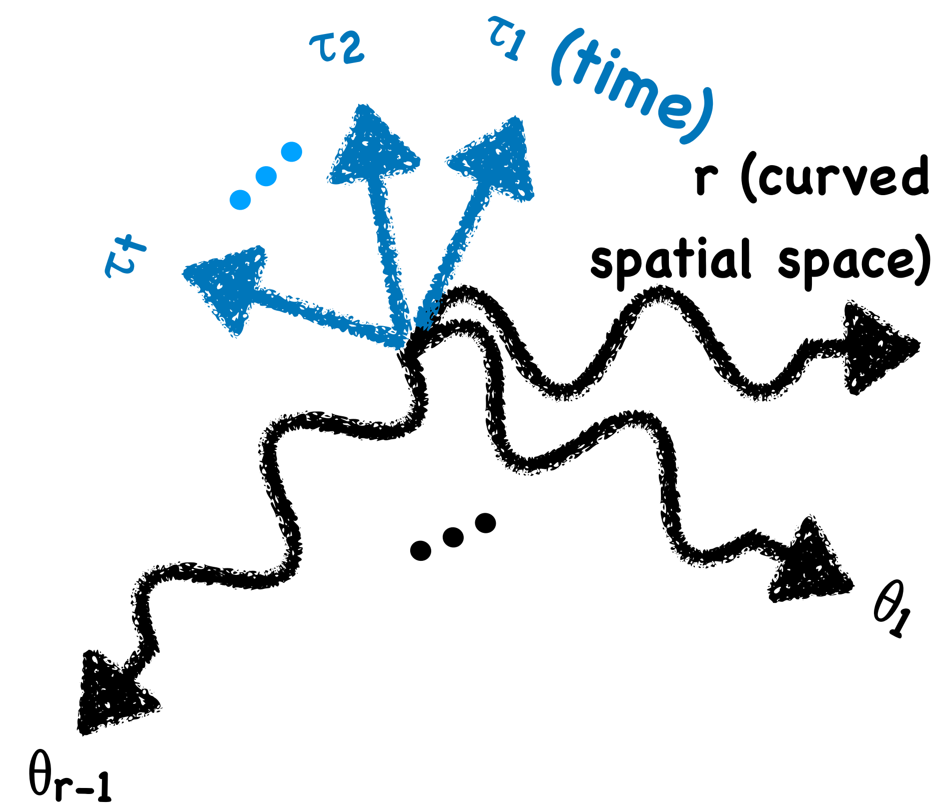
(Dated: 24 February 2023)

In this document, we introduce a novel formalism for any field theory and apply it to the effective field theories of large-scale structure. The new formalism is based on *functors of actions* composing those theories. This new formalism predicts the *actionic* fields. We discuss our findings in a *cosmological gravitology* framework. We present these results with a cosmological inference approach and give guidelines on how we can choose the best candidate between those models with some latest understanding of model selection using Bayesian inference.

A (Dτ,Dx)-manifold with NPCF of Nt-objects

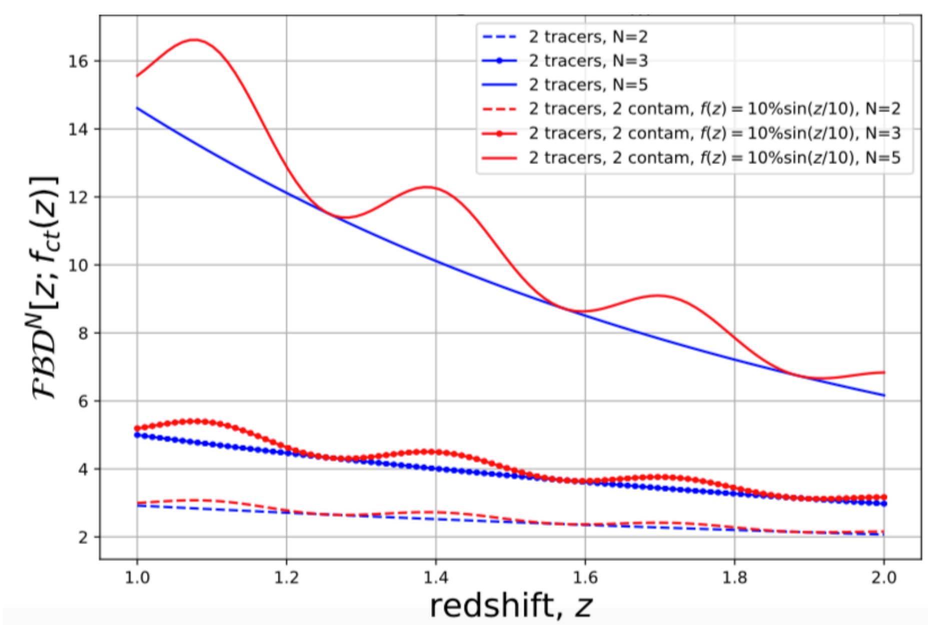


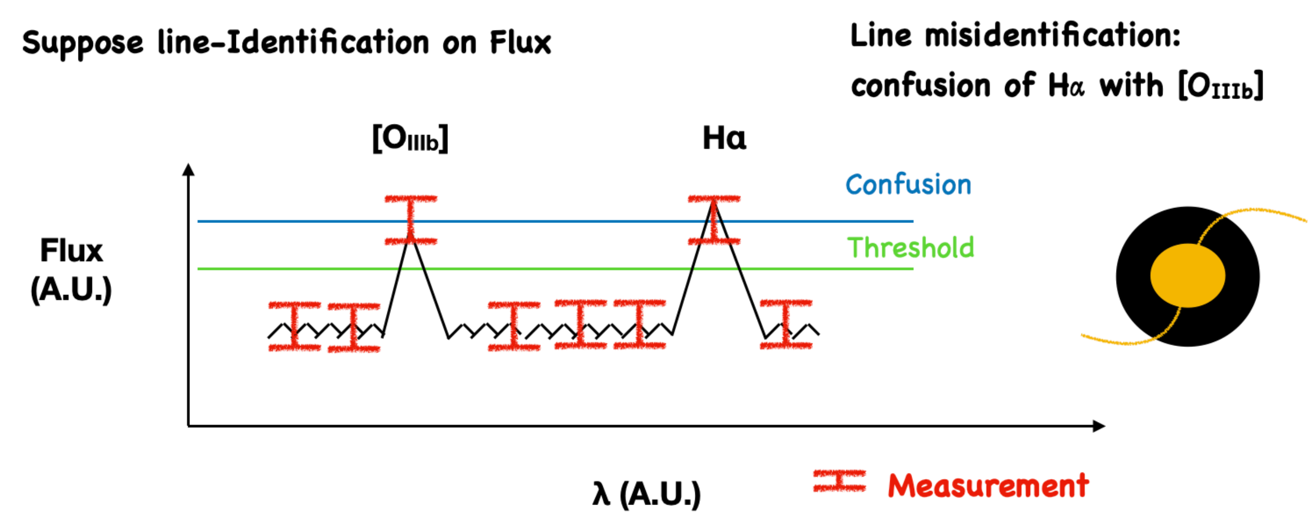
- **extra dimensions**
- **contaminated targets**
- Applied to **Cosmic Structures**:
  - **Large** scale structure (**Astronomical**)
  - **Small** scale structure (**Quantum**)



$$\mathcal{FBD}(\vec{\tau},\vec{x}) = \sum_{t=1}^{N_t^{\rm OLSS}} \left\{ \left[1-\sum_{c=1}^{N_{ct}} f_{ct}(\vec{\tau},\vec{x})\right] b_t(\vec{\tau},\vec{x}) D_t(\vec{\tau},\vec{x}) + \sum_{c=1}^{N_{ct}} |\vec{\gamma}_{ct}|^{D_x}(\vec{\tau},\vec{x}) f_{ct}(\vec{\tau},\vec{x}) b_{ct}(\vec{\tau},\vec{x}) D_{ct}(\vec{\tau},\vec{x}) \right\} \quad (3.19)$$

Functional Biased Growth





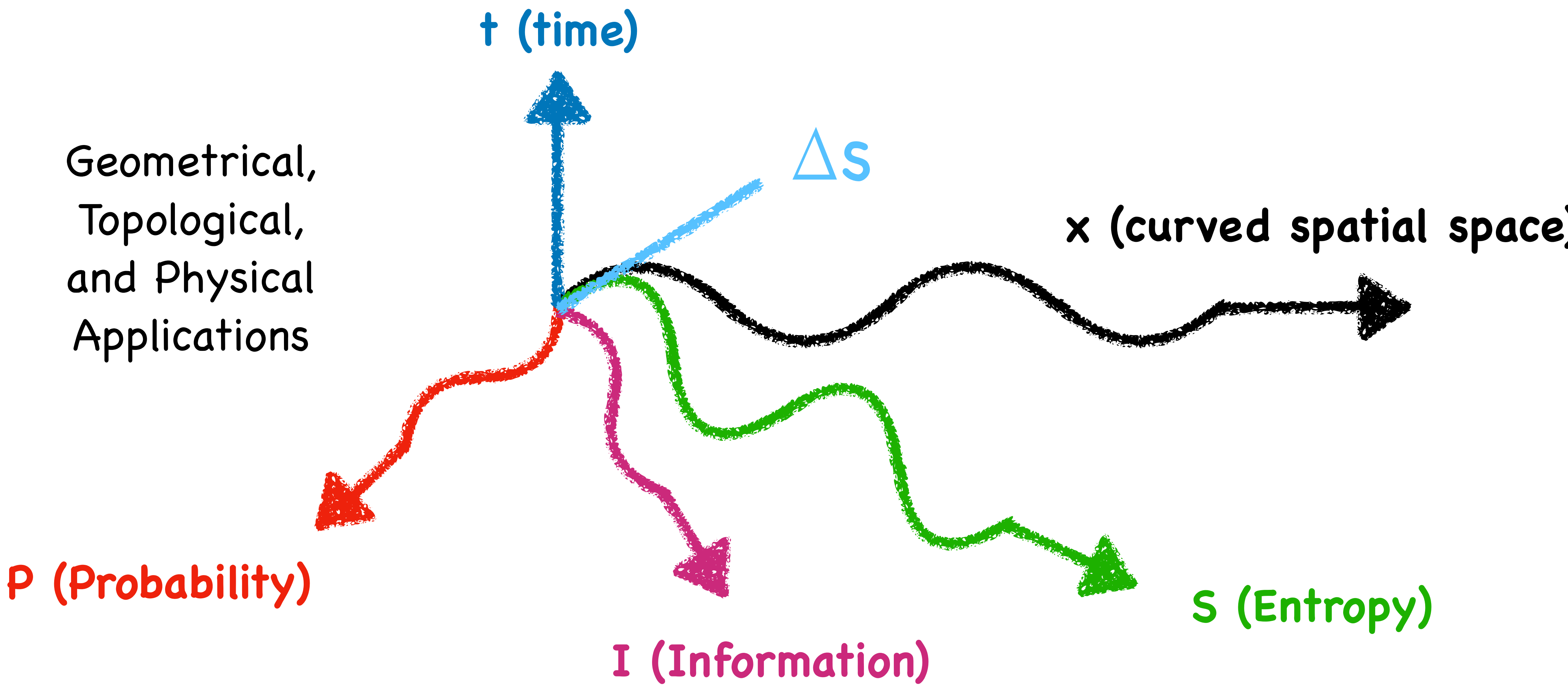
… Further Simulations …

PN (2209.07472) IJGMMP Jan 2026

Advanced Manifold Metric Pairs

$$ds^{U+L} = \mathcal{F}^{\nu_{\vec{P}}\nu_{\vec{I}}\nu_{\vec{S}}}_{\mu_t\mu_{\vec{x}}}[c, f(c)]dc^{\mu_t}dc^{\mu_{\vec{x}}}dc_{\nu_{\vec{P}}}dc_{\nu_{\vec{I}}}dc_{\nu_{\vec{S}}}$$

$$c = \left[t, \vec{x}, \vec{P}, \vec{I}, \vec{S}\right]$$



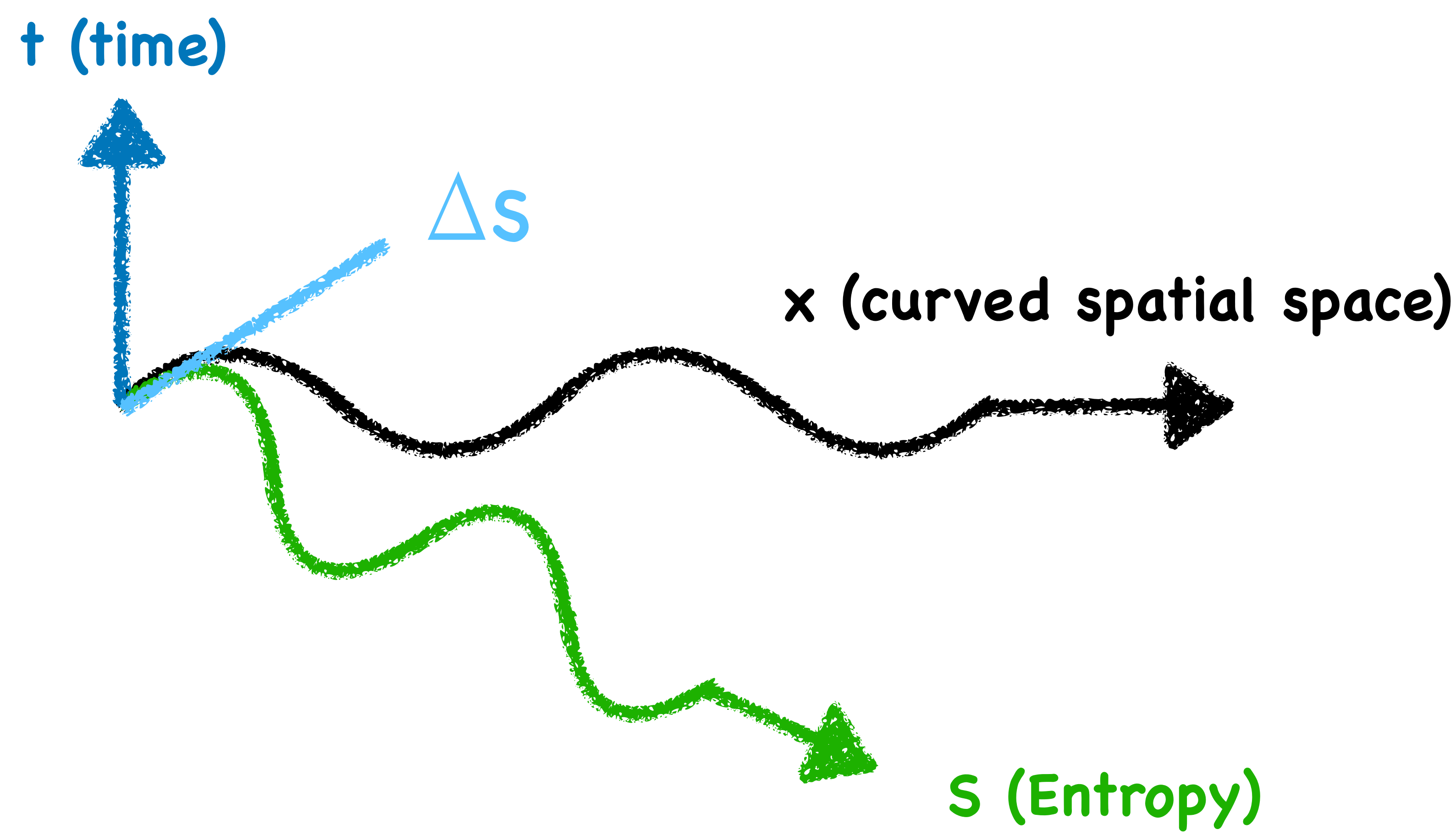
… Further Simulations …

Published in MDPI Mathematics Journal
Visualisation via macos, keynote and zoom

Advanced Manifold Metric Pairs

$$ds^{U+L} = \mathcal{F}^{\nu_{\vec{P}}\nu_{\vec{I}}\nu_{\vec{S}}}_{\mu_t\mu_{\vec{x}}}[c, f(c)]dc^{\mu_t}dc^{\mu_{\vec{x}}}dc_{\nu_{\vec{P}}}dc_{\nu_{\vec{I}}}dc_{\nu_{\vec{S}}}$$

$$c = \left[t, \vec{x}, \cancel{\vec{P}}, \cancel{\vec{I}}, \vec{S}\right]$$



Published in MDPI Mathematics Journal
Visualisation via macos, keynote and zoom

A temporal-spatial-entropic K-curvature

$$ds^2 = -c^2 dt^2 + a(t)^2(dx^2 + dy^2 + dz^2) + f_S(t,S)^2 dS^2$$

Considering its Kretschmann scalar

$$K = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta}$$

Compare txyzS to FLRW, with the ratio

$$\frac{K_{\text{txyzS}}}{K_{\text{FLRW}}} = 1 + \frac{\frac{4}{c^4}(\beta S)^4 + \frac{12}{c^4}\left(\frac{\dot{a}}{a}\right)^2 (\beta S)^2}{\frac{12}{c^4}\left(\frac{\ddot{a}}{a}\right)^2 + \frac{12}{c^4}\left(\frac{\dot{a}}{a}\right)^4}$$

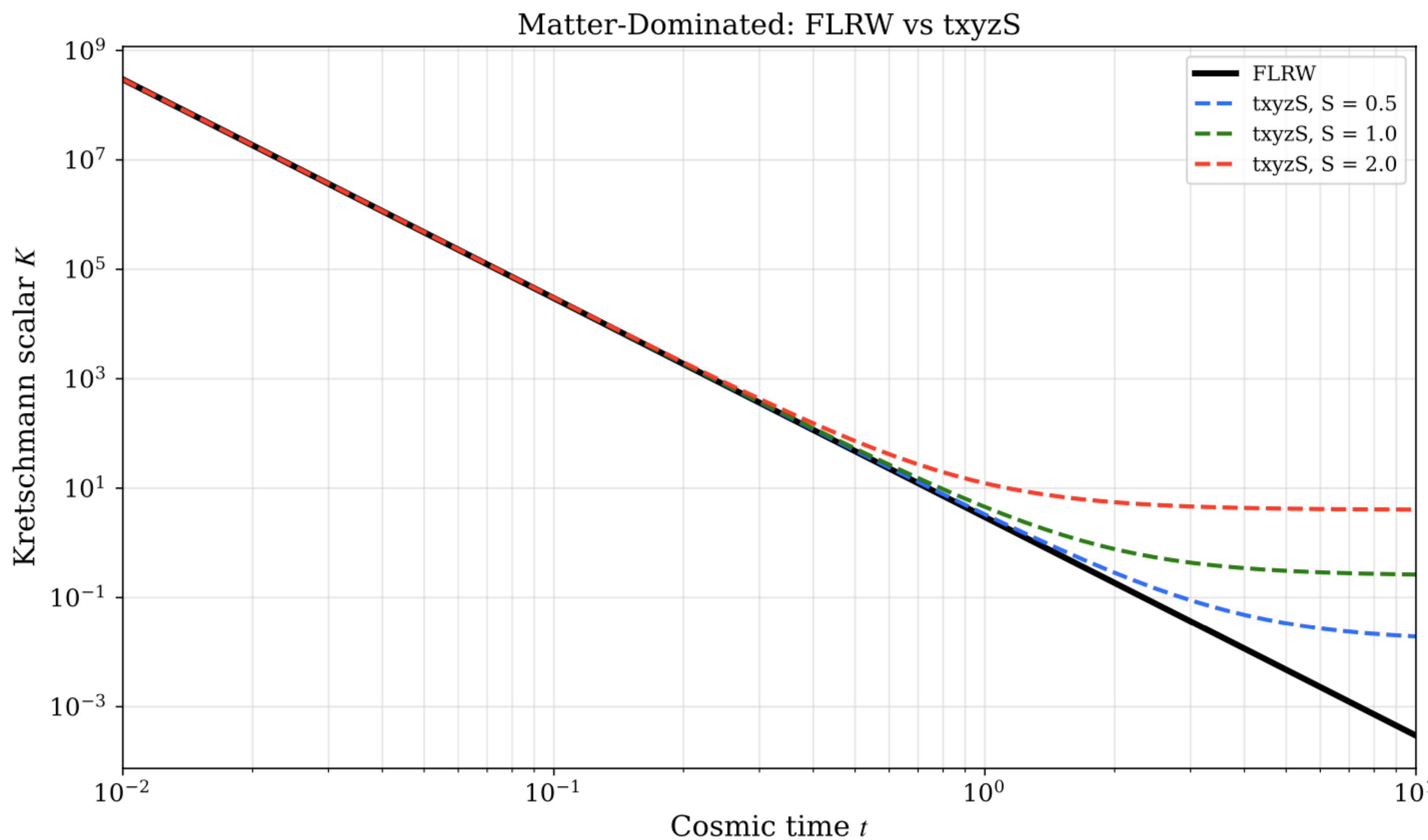


Figure 7: Comparison of Kretschmann scalar: FLRW vs txyzS (matter-dominated). The entropy dimension introduces an additional constant term $4(\beta S)^4$ and a coupling term $12H^2(\beta S)^2$, both increasing K relative to FLRW.

Predictions to be constrained by

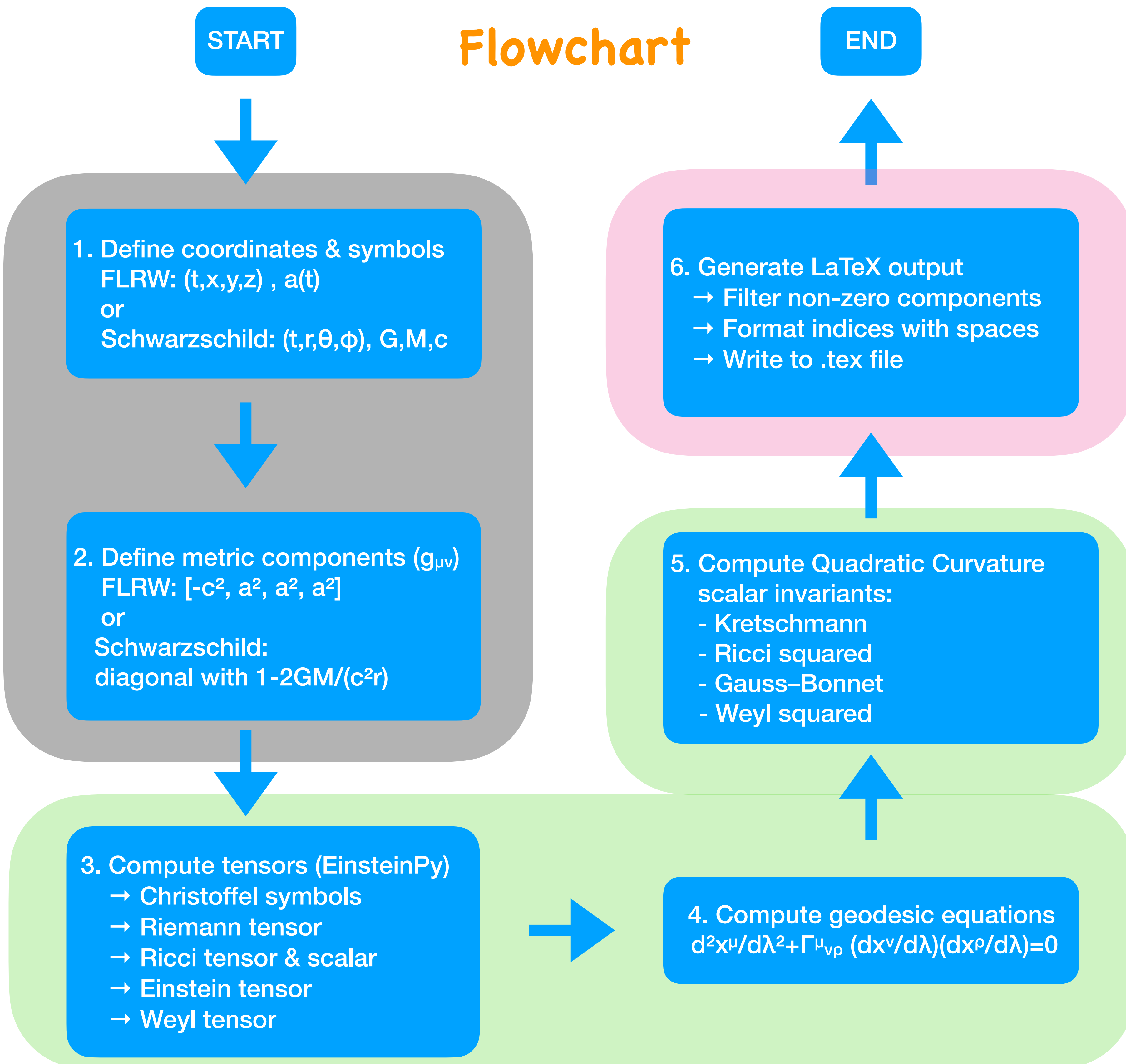
SnIa

LSS

CMB

Submitted EinsteinKPy2Tex code

EinsteinKPy2Tex code



EinsteinKPy2Tex code

Easy to install

Flexible to expand and add modules

Efficient documentation and examples

Appropriate for educational and research purposes

EinsteinKPy2Tex: A Python Tool for Computing and Exporting General Relativity Tensors to LaTeX

Pierros Ntelis[†,‡]

[†]School of Physics, Harbin Institute of Technology, Harbin 150001, People's Republic of China
[‡]Institute of Theoretical Physics, National University of Uzbekistan, Tashkent 100174, Uzbekistan

June 7, 2026; conceived: 5 June 2026

Abstract

We present EinsteinKPy2Tex, an open-source Python package that computes fundamental geometric quantities for the Friedmann-Lemaître-Robertson-Walker (FLRW) and Schwarzschild metrics using the EinsteinPy library. The code automatically calculates Christoffel symbols, Riemann curvature tensor, Ricci tensor, Ricci scalar, Einstein tensor, geodesic equations, and Kretschmann scalar. Results are exported as professionally formatted LaTeX documents, facilitating direct inclusion in research papers and educational materials.

The FLRW implementation is essential for cosmology, enabling the study of the universe's large-scale structure and dynamics. The Schwarzschild implementation is fundamental for black hole physics, describing the spacetime geometry around non-rotating compact objects. The code includes built-in citations for the original Einstein paper (1915) and the EinsteinPy library (Ribeiro et al. 2020), ensuring proper academic attribution.

EinsteinKPy2Tex is designed for educators, students, and researchers needing rapid, accurate tensor calculations with publication-ready output. The software is available at GitHub under an MIT license.

Keywords: General Relativity ; Schwarzschild metric ; FLRW metric ; Kretschmann scalar ; Geodesic equations ; Tensor calculus ; LaTeX export ; Symbolic computation ; EinsteinPy ; EinsteinKPy2Tex ; Cosmology ; Black hole physics

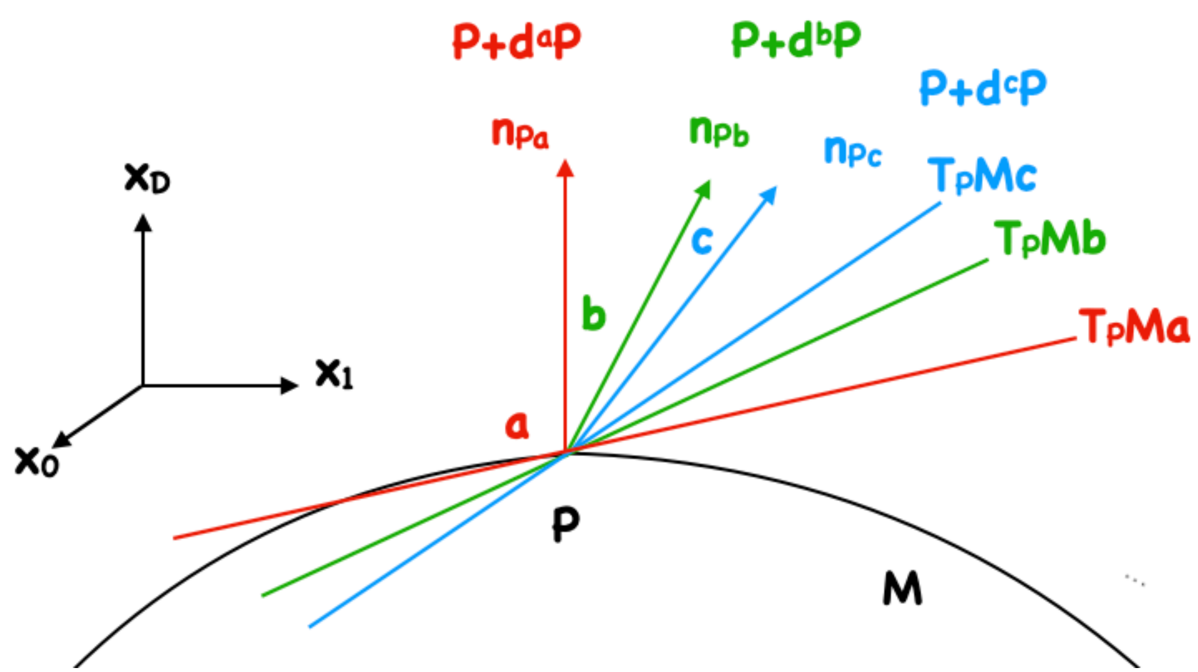
Contents

Advancing Tensor Theories

the generic tensor, T, is simply the map
between the generic combination of vectors, V,
to the generic number set, G, written as

$$\mathcal{T} : \mathcal{V} \to \mathbb{G}$$

Geometrical and Topological interpretation is done using fractional calculus:



Published in Symmetry: Advances in Algebraic Geometry and Topology
Visualisation via MacOS, Python, and Grok

Advancing Tensor Theories

In the literature, there are tensor such as

$$T = T^{\mu_1 \mu_2}{}_{\nu_1 \nu_2}$$

Note that each index has its own index

=> NEW IDEA !

The index of the index … of the index

Generic index $$\mathcal{I} = \ell_{k_{\dots_{j_i}}}$$

=> Generic tensor ! $$T = T^{\ell_{k_{\dots_{j_i}}}}{}_{\alpha_{\beta_{\dots_{\psi_\omega}}}}$$

New mathematical object!

A Foundational Analysis of Kernel-based Calculus (AFAKEC)

Kernel Based Derivative

$$D_x^{F,\alpha} f(x) = \lim_{\Delta x \to 0} \frac{f\big(x + \Delta x \, F(x,\alpha)\big) - f(x)}{\Delta x}$$

$$D_x^{F,\alpha} f(x) = F(x,\alpha) f'(x)$$

Kernel Based Chain Rule

$$D_x^{F,\alpha}\Big[f\big(g(x)\big)\Big] = D_g^{F,\alpha}\Big[f(g)\Big] \cdot D_x^{F,\alpha}\big[g(x)\big] \cdot F^{-1}\big(g(x),\alpha\big)$$

Kernel Based Integral

$$I_x^{F,\alpha} f(x) = \int \mathrm{d}_{F,\alpha} x \; f(x) = \int \frac{\mathrm{d}x}{F(x,\alpha)} \; f(x)$$

Fundamental Theorem of Local Kernel-based Calculus (FTLKEC)

$$\int_{x_{\min}}^{x_{\max}} \frac{D_x^{F,\alpha} f(x)}{F(x,\alpha)} \, \mathrm{d}x = f(x_{\max}) - f(x_{\min})$$

Proof:

$$\int_{x_{\min}}^{x_{\max}} \frac{D_x^{F,\alpha} f(x)}{F(x,\alpha)} \, \mathrm{d}x = \int_{x_{\min}}^{x_{\max}} \frac{F(x,\alpha) D_x f(x)}{F(x,\alpha)} \, \mathrm{d}x = \int_{x_{\min}}^{x_{\max}} f'(x) \, \mathrm{d}x = f(x_{\max}) - f(x_{\min})$$

Identity operator of Local Kernel-based Calculus

$$I_t^{F,\alpha,x_0,x}\left(D_t^{F,\alpha} f(t)\right) = f(x) - f(x_0)$$

Proof:

$$I_t^{F,\alpha,x_0,x}\left(D_t^{F,\alpha} f(t)\right) = \int_{x_0}^{x} \frac{D_t^{F,\alpha} f(t)}{F(t,\alpha)}\, dt = \int_{x_0}^{x} \frac{f'(t)\, F(t,\alpha)}{F(t,\alpha)}\, dt = \int_{x_0}^{x} f'(t)\, dt = f(x) - f(x_0)$$

Remark 21. *Conversely, differentiating a local kernel-based integral returns the original function:*

$$D_x^{F,\alpha}\left(I_x^{F,\alpha} f(x)\right) = D_x^{F,\alpha}\left(\int \frac{f(x)}{F(x,\alpha)}\, \mathrm{d}x\right) = \frac{d}{\mathrm{d}x}\left(\int \frac{f(x)}{F(x,\alpha)}\, \mathrm{d}x\right) \cdot F(x,\alpha) \tag{142}$$

$$= \frac{f(x)}{F(x,\alpha)} \cdot F(x,\alpha) = f(x). \tag{143}$$

Thus $D_x^{F,\alpha} I_x^{F,\alpha} = I$ on the space of differentiable functions where the relevant integrals exist and $F(x,\alpha) \neq 0$. For more general function classes (e.g., merely integrable functions), the identity holds only in the almost-everywhere sense or under additional regularity assumptions.

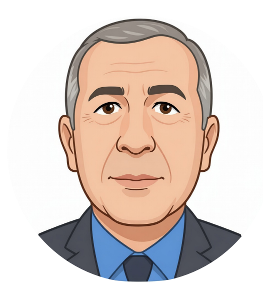
Fractional Calculus in Cosmology

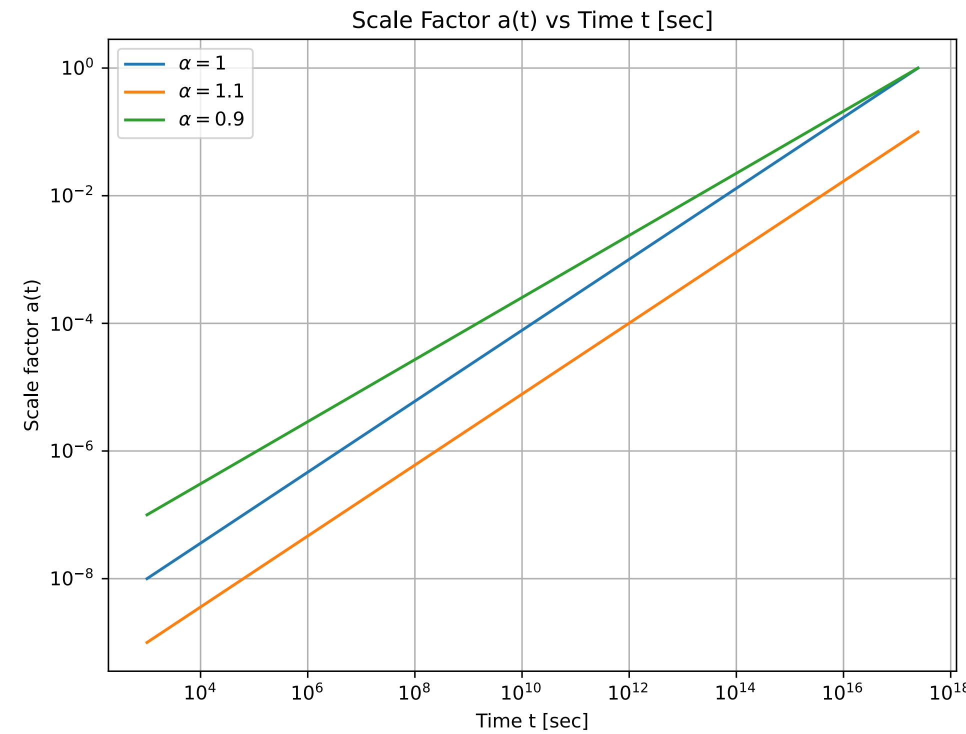
with the N. Uzbekistan University Group

- Cosmic applications with Pakhlavon Yovkochev

- Foundational aspects with Bakhodir Shodikulov

Supervised by B.J. Ahmedov, and PN





Fiber bundle: base space B, total space E, fiber F, projections π.

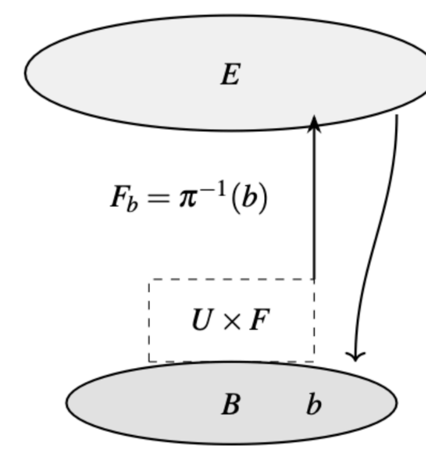
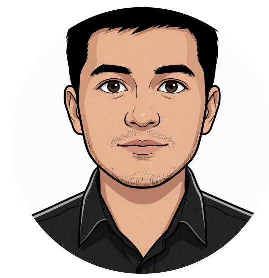


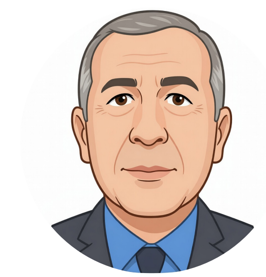
Figure 1: Schematic of a Fibre bundle (E, B, π, F).

Preliminary results

Chaotic motion and power spectral density in Schwarzschild Bertotti-Robinson black hole spacetime

by Y. Xu, U. Uktamov, PN, A. Abdujabbarov, B.J. Ahmedov, C. Yuan


$$ds^2 = \frac{1}{R^2}\left[-Qdt^2 + \frac{dr^2}{Q} + r^2\left(\frac{d\theta^2}{P} + P\sin^2\theta d\phi^2\right)\right]$$

$$P = 1 + B^2M^2\cos^2\theta$$

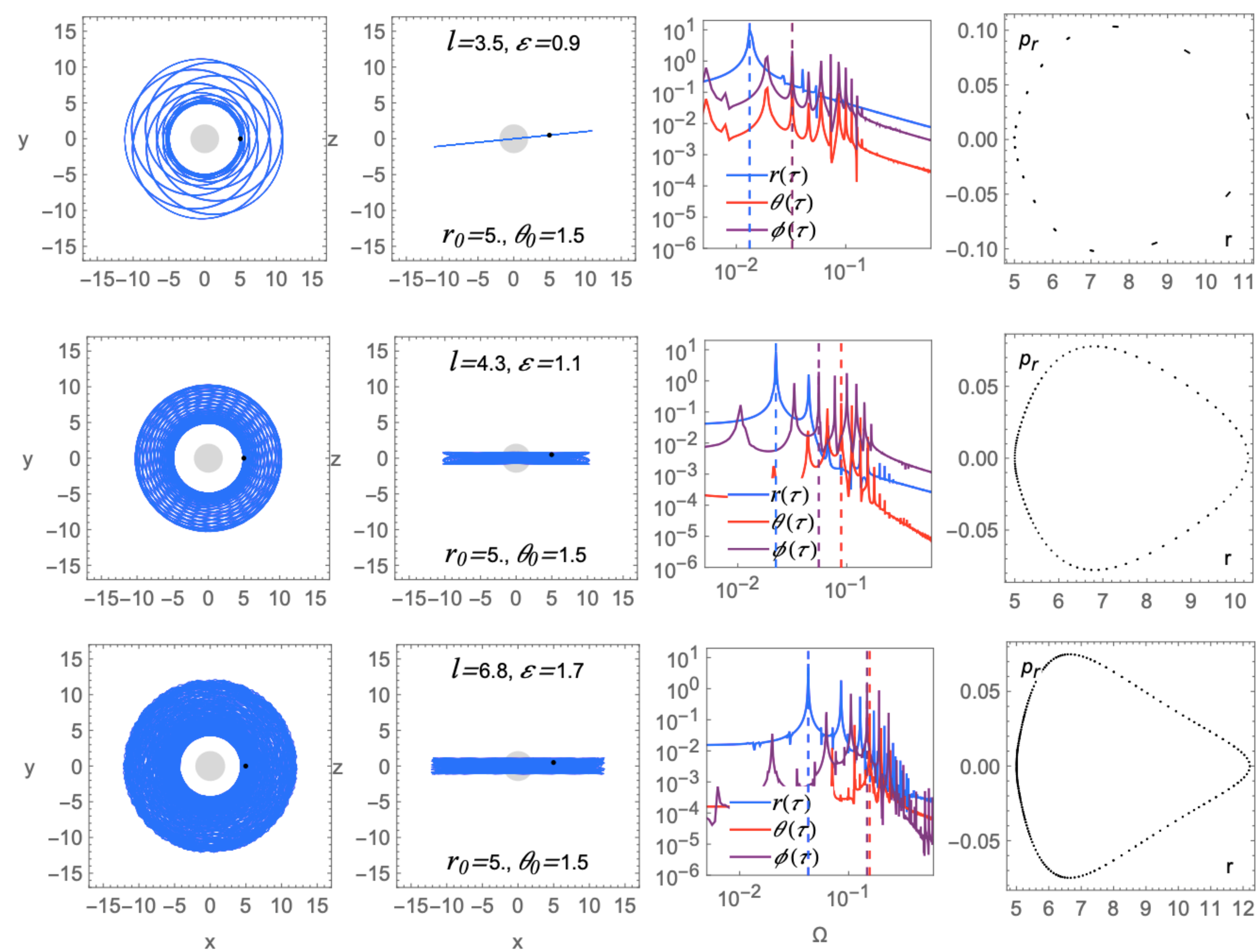
$$Q = \left(1 + B^2r^2\right)\left(1 - B^2M^2 - \frac{2M}{r}\right)$$

$$R^2 = 1 + B^2\left[r^2\sin^2\theta + \left(2Mr + B^2M^2r^2\right)\cos^2\theta\right]$$

M = mass field of BH

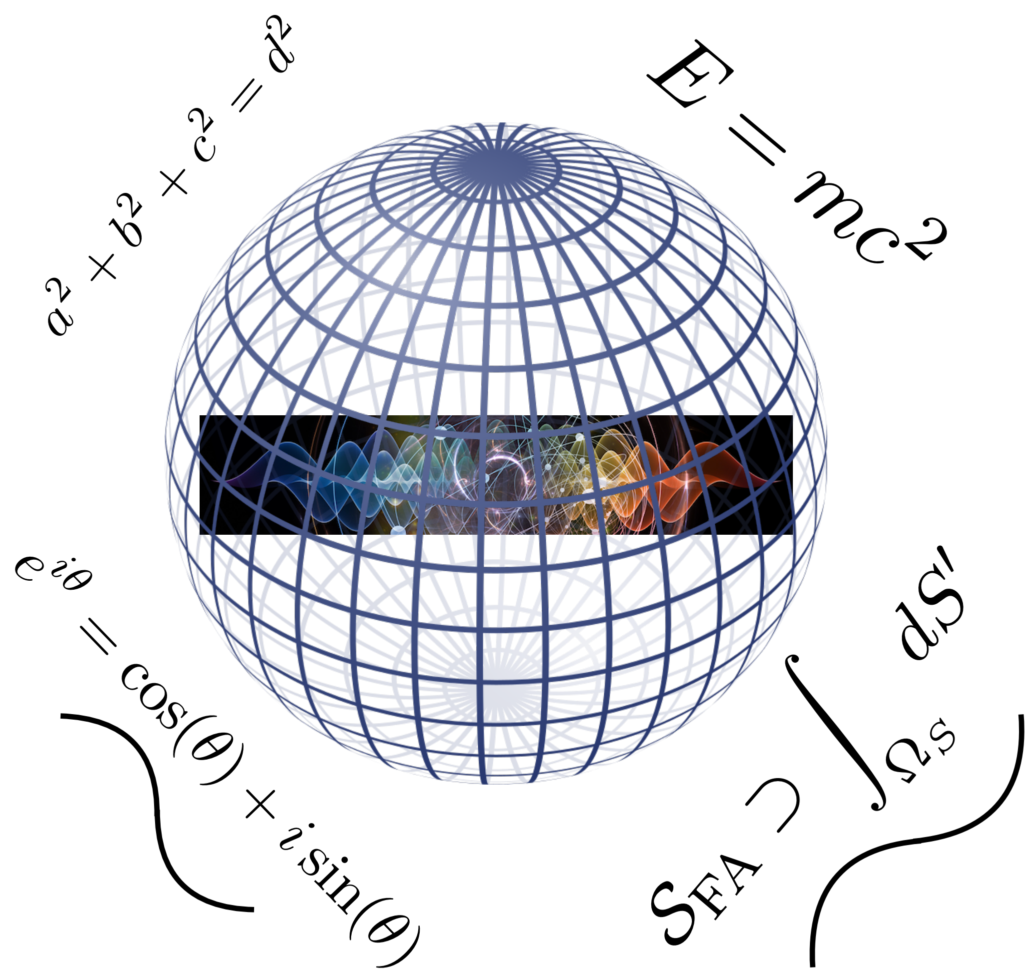
B = Magnetic Field around BH distorting the spacetime

Charged Test particle stabilises its orbit, as Magnetic Field increases



Science Popularisation !

Universions



YouTube

Analogue

bilibili

Thank you

谢谢

Xiè xiè

! :)

Subscribe to support Art & Science ! :)

Conclusions and Outlook

- Mathematical and Observational evidence for
  - Functors of actions theories and predictions of actionions
  - Quintessence ($\varphi\Lambda$CDM)
- Dynamical analysis on
  - Standard cosmology (ΛCDM)
  - Quintessence ($\varphi\Lambda$CDM) (Complex and Simple DAC)
  - polyΛCDM model
- Advancing Manifold-Metric Pairs
  - Functional (U,L)-rank tensor metric
  - Restructuring extra dimensions with t, x, P, I, S
- A (Dτ,Dx)-manifold with NPCF of Nt-objects
- Advancing Tensor Theories
  - Generic Indices, Generic Tensors
  - Generic Tensors connected with Fractional Calculus
  - Fractional Calculus in Cosmology

Thank you

谢谢

Xiè xiè

Ευχαριστώ

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